

Self-Regenerating Zwitterionic Hydrogel Coating Modified by Metal-Organic Framework for Integrated Anti-Biofouling and Anti-Corrosion

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Hydrogel coating has a certain ability to inhibit marine biofouling. However, owing to their poor mechanical properties and short service life, they have not found solid applications in marine protection. Herein, a novel self-regenerating zwitterionic hydrogel coating composed of a self-crosslinking degradable polymer with hydrolysis-induced zwitterions and corrosion inhibitor benzotriazole (BTA) loaded metal-organic framework (MOF) is reported, which exhibits both anti-biofouling and anti-corrosion performance. The polymer is prepared by copolymerization of 2-methylene-1,3-dioxepane (MDO) and vinyl chloroacetate (VCA) followed by grafting with 2-(trimethoxysilyl) propyl dimethylamino propionate (TPD). It can self-crosslink or crosslink with BTA-loaded MOF forming a network, and even chemically link the substrate creating a robust interface. Accordingly, the hydrogel coating exhibits good mechanical properties and adhesion. Because the zwitterionic surface with antifouling ability can be continuously renewed by the degradation and hydrolysis of the polymer, the coating has high resistance to bacteria and diatom in seawater. The degradation and hydrolysis also enable the release of BTA, significantly inhibiting surface corrosion. This study represents an important step toward the integration of hydrogel into anti-biofouling and anti-corrosion coating.

tributyltin due to its negative environmental impacts, great effort has been made to develop environment-friendly anti-biofouling coatings.^[7,8] For this purpose, fouling release coatings (FRC)^[9] and self-polishing copolymer (SPC) coatings^[10] have been developed. Owing to their good anti-biofouling properties and environmental friendliness, hydrogel coatings received considerable attention.^[11,12] Particularly, zwitterionic hydrogels with strong and stable hydration layers demonstrate excellent resistance to biofouling.^[13]

However, hydrogel coatings suffer from poor adhesion because of the presence of a considerable amount of water at the interface.^[14] To improve the adhesion, some chemical modification techniques have been developed.^[12,15–18] Unfortunately, the hydrogel coatings do not have self-regeneration ability, so they gradually lose their anti-biofouling ability with the adhesion of inorganic substances or dead bacteria.^[19] On the other hand, hydrogel coatings generally exhibit poor mechanical properties in marine environments due to

their swelling.^[20] They cannot also create an additional physical barrier to effectively prevent corrosive substances from penetrating, failing to impede the corrosion.^[21]

It is well established that dynamic surfaces can significantly reduce the contact time and force between organisms and surfaces, improving their antifouling performance.^[22,23] On the other hand, metal-organic frameworks (MOFs) characterized by metal cations formed clusters with organic linkers as ligands have received much attention recently because they have porous nanostructures, adaptable morphology, and high specific surface area.^[24–26] It was reported that UiO-66, one of the MOFs, exhibited anti-corrosive properties due to the inhibitory effect of the metal ions at the cathodic site and the organic ligands at the anodic site.^[27,28] Recently, some attempts have been made to utilize MOFs as nanocontainers to carry inhibitors such as benzotriazole (BTA) to effectively block metal corrosion.^[29,30] In the present study, we report a self-regenerating zwitterionic hydrogel coating (MUX) composed of a degradable polymer (PMVSx) and benzotriazole (BTA)-loaded UiO-66, where the polymer was prepared from copolymer (PMV) of 2-methylene-1,3-dioxepane (MDO) with vinyl chloroacetate (VCA) grafted with 2-(trimethoxysilyl)

1. Introduction

Marine biofouling is becoming a global issue.^[1,2] The fouling organisms on marine vessels promote surface deterioration, damage propellers, and increase drag, thereby leading to high fuel consumption and excessive maintenance costs.^[3,4] Using anti-biofouling coatings is a highly effective and practical solution to combat biofouling.^[5,6] Given the stringent regulations governing the utilization of biocide agents, particularly the prohibition of

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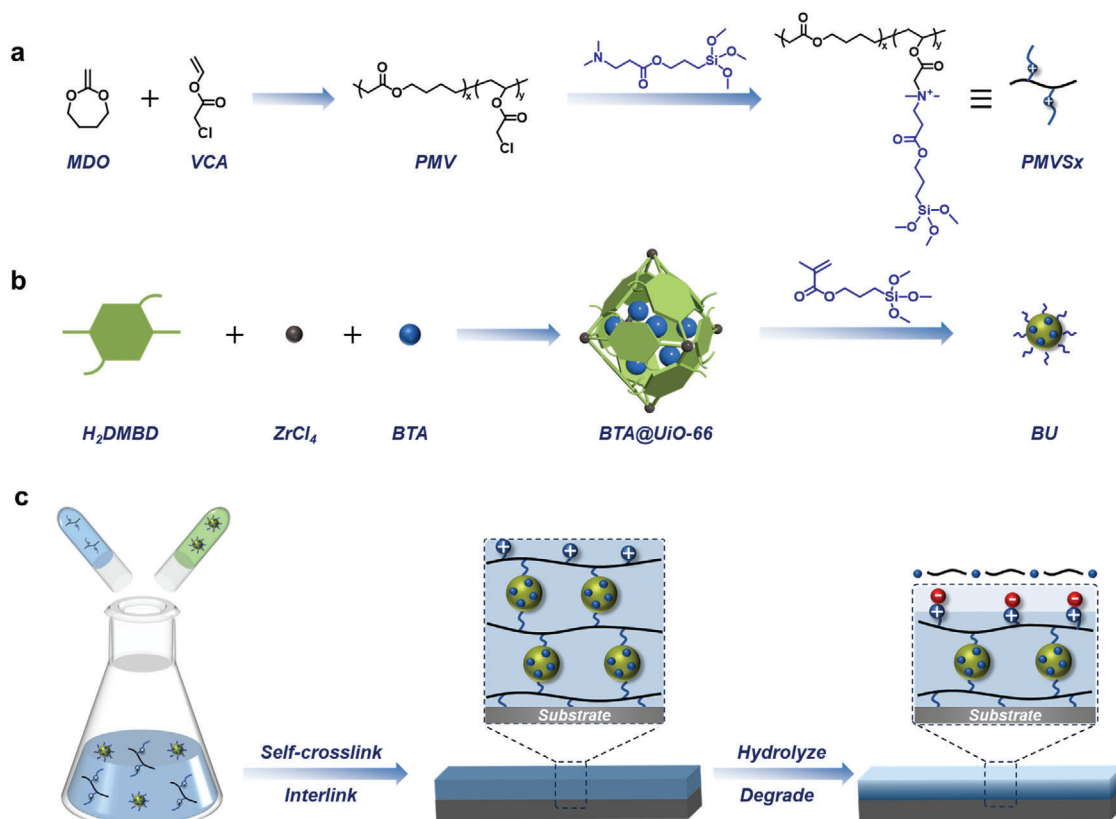


Figure 1. Illustration of Preparation of MUx. a) Synthesis of PMVSx. b) Synthesis of BU. c) Preparation of MUx. The hydrolyzation and degradation of the coating generate a renewable zwitterionic surface and release MOF. The loaded BTA can be released by disintegration of MOF.

propyl dimethylamino propionate (TPD) acting as zwitterion precursor and offering cross-linking sites for silane-modified MOFs. The hydrolyzation and degradation of the polymer on the coating surface in seawater generate a renewable or zwitterionic dynamic surface for anti-biofouling while BTA is released to inhibit corrosion. The wettability, degradation, anti-biofouling, and anti-corrosion performances of the coating were systematically investigated.

2. Results and Discussion

2.1. Preparation of PMVSx

It is well known that the poor mechanical properties and short service life prevent hydrogel coatings from application in anti-biofouling. Therefore, a self-regenerating zwitterionic hydrogel coating with BTA-loaded MOF was fabricated herein. Due to the degradation of polyester segments and renewal of zwitterion on the surface,^[23,31] the coating showed high resistance to marine biofouling. Furthermore, the incorporation of silane-modified MOF improved its mechanical properties and adhesion to the substrate. The degradation and disintegration of UiO-66 in alkaline and saline environments enabled the release of the corrosion inhibitor (BTA) for corrosion protection.

MUx was fabricated through silanol condensation between PMVSx and BU. PMV was synthesized via radical ring-opening polymerization of MDO and VCA (Figure 1a). The ¹HMR spec-

tra of the product show peaks at 4.17 and 2.65 ppm assigned to the methylene of polyester, but the peaks at 4.71 and 4.99 ppm assigned to C=C in VCA are absent (Figure S1, Supporting Information). On the other hand, FTIR spectra show the band at 1060 cm⁻¹ assigned to C–O stretching vibration in polyester, whereas the band at 1650 cm⁻¹ assigned to C=C stretching vibration disappears. The facts indicate the successful synthesis of PMV (Figure S2, Supporting Information). The monomer conversion of MDO was close to that of VCA (Figure 2a), indicating that the two monomers have a reactivity ratio close to each other and the copolymer is random. Namely, the ester groups are randomly distributed along the chain. Thus, PMV can degrade into small molecules, reflecting that *M_n* decreased from 2500 to 500 g mol⁻¹ (Figure 2b). To introduce zwitterionic precursors and cross-linkable groups, PMVSx was prepared via a quaternization reaction by grafting TPD to PMV. The ¹H NMR spectra of TPD and PMVSx shown in Figure 2c and Figure S3 (Supporting Information) indicated their successful synthesis.

BU was prepared by encapsulating BTA in a one-pot hydrothermal procedure and its surface was modified with silane coupling agents 3-(methacryloxypropyl) trimethoxysilane (KH570) (Figure 1b). The resulting MUx hydrogel coating can hydrolyze and degrade in seawater to form a zwitterionic dynamic surface improving the static anti-biofouling performance,^[20,31] while BTA loaded by BU is released to inhibit corrosion (Figure 1c). Furthermore, the alkaline and saline environments can promote the disintegration of BU and release of BTA and hence the formation

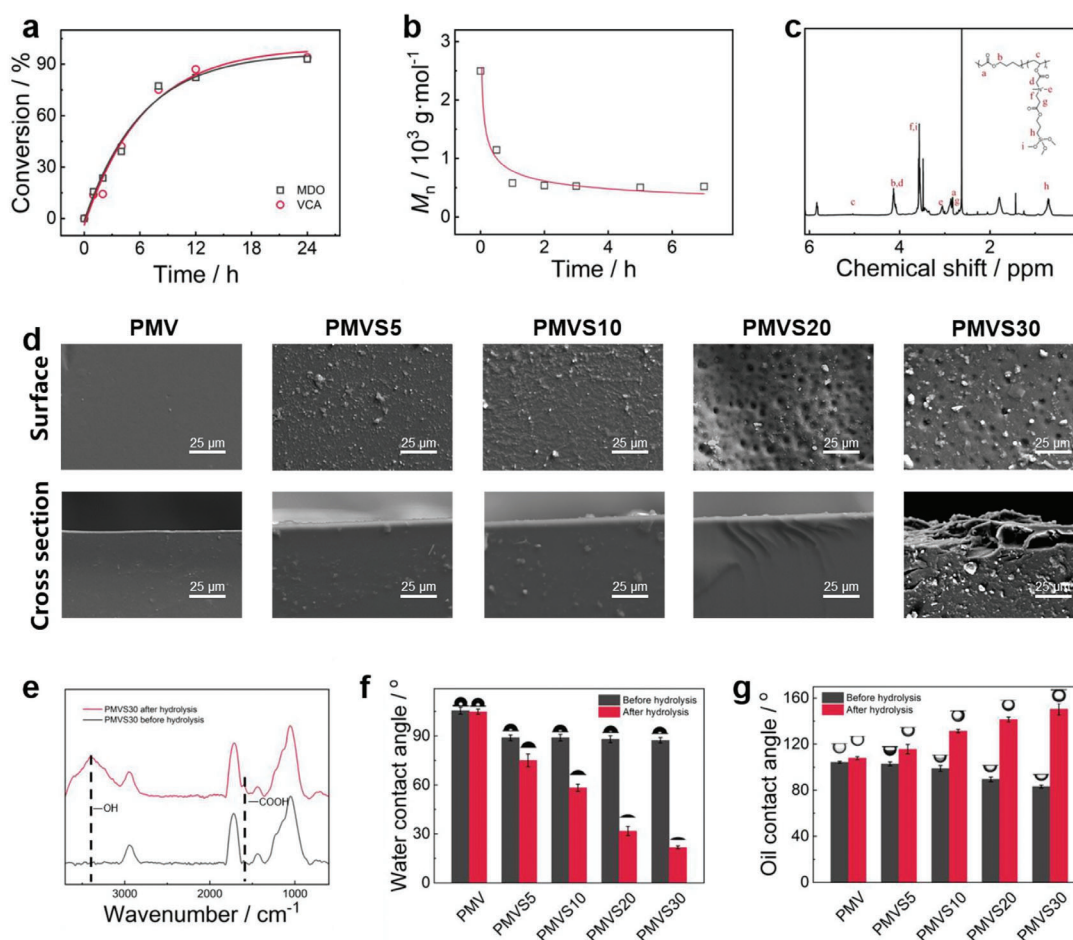


Figure 2. Characterization of PMVSx. a) Time dependence of monomer conversion in MDO-VCA copolymerization. b) Changes in the number average molecular weight (M_n) of PMV during accelerated degradation. c) ^1H NMR spectrum of PMVSx in CDCl_3 . d) SEM images of the cross-section and surface of PMVSx coating after hydrolysis. e) ATR-IR spectra of PMVS30. f) Water contact angle (WCA) in air and g) Oil contact angle (OCA) under water for PMVSx before and after hydrolysis.

of a passivation layer to isolate the corrosive medium, integrating anti-biofouling and anti-corrosion.^[32,33] The hydrolysis does not occur on the bottom layer but on the top layer since the coating has a Janus structure with a self-regenerating zwitterionic layer on top and an unhydrolyzed hydrophobic layer on the bottom. As a result, the coating shows improved anti-biofouling performance and high adhesion to the substrate.^[18]

Coatings with different TPD grafting densities were prepared to regulate the hydrolysis and swelling degree. Scanning electron microscopy (SEM) images demonstrate that PMV had a non-porous and dense microstructure on the surface and cross-section (Figure 2d). PMVS5 and PMVS10 had relatively flat surfaces with a uniform and dense structure due to the low TPD grafting density. However, PMVS20 showed a surface structure with a large number of pores, and PMVS30 had a porous structure on both its surface and cross-section. Attenuated total reflection infrared (ATR-IR) spectra show new bands at 3387 and 1576 cm^{-1} assigned to hydroxyl and carboxyl groups, respectively, indicating hydrolysis of zwitterionic precursors (Figure 2e).

The hydrophilicity and oleophobicity of the hydrogel coating surface were characterized by contact angle measurements

(Figure 2f,g). PMVSx had a lower contact angle than PMV because the quaternarization increased the hydrophilicity of the former. As TPD density increased, the WCA of PMVSx slightly decreased. More decrease was observed after its hydrolysis because more hydrophilic zwitterions were generated. Similarly, the initial underwater OCA of PMVSx slightly decreased with oleophilic silane coupling agent content, whereas the hydrolyzed PMVSx exhibited a notably increased underwater OCA due to the oleophobicity of the zwitterions. A relatively high TPD grafting density facilitates the generation of hydrophilic zwitterions upon hydrolysis, which enables the coatings to swell and form a hydrogel in aqueous environments.^[18] Considering that PMVS30 hydrolyzes too quickly and swells too severely to be used, we chose PMVS20 for all the tests.

2.2. Preparation of BU and MUX Coatings

UiO-66 particles and UiO-66 particles with BTA (BU) were prepared using a procedure reported before.^[34] As shown in Figure 3a,b,d,e, both UiO-66 and BU exhibited octahedral

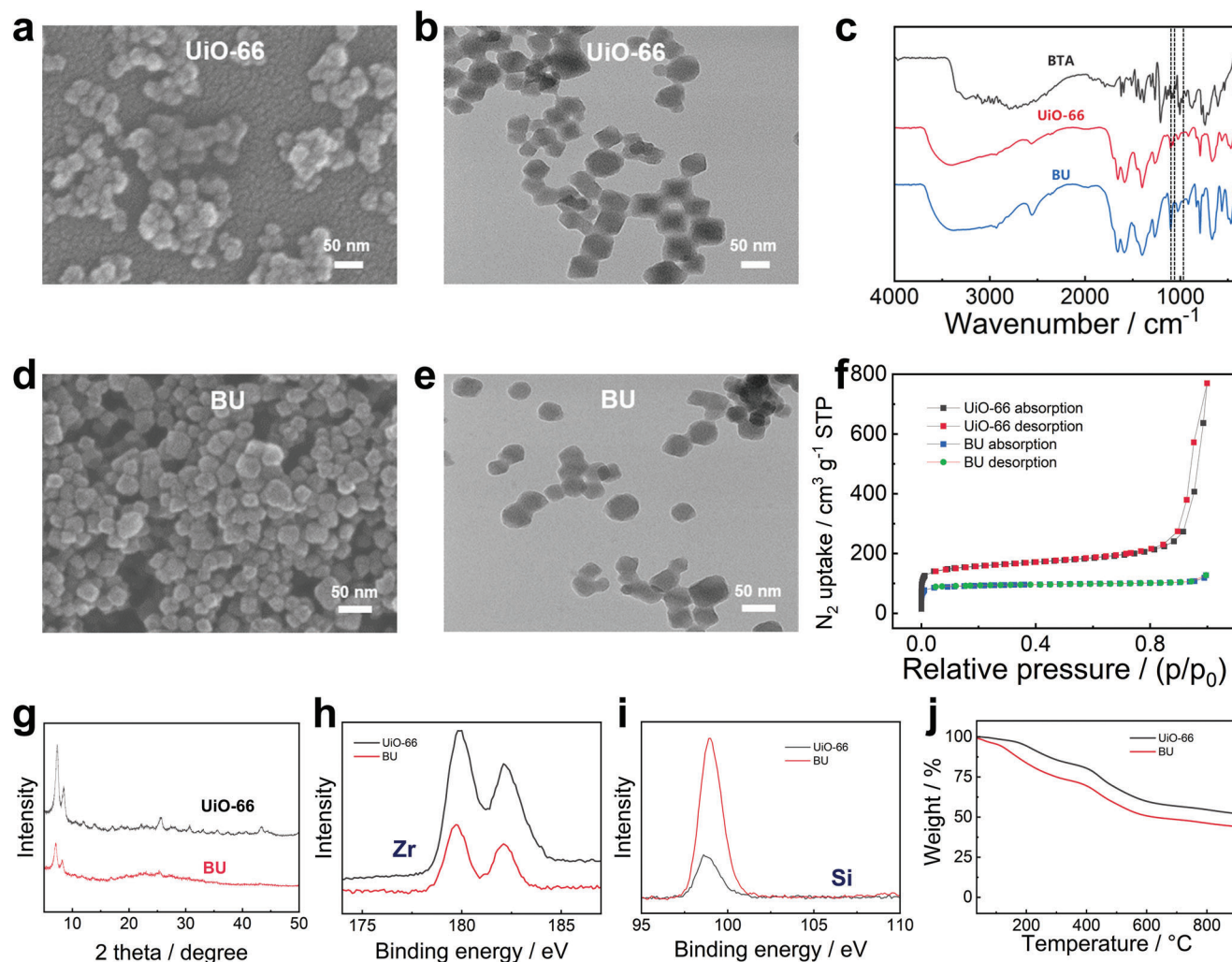


Figure 3. Characterizations of MOF (UiO-66) and BU. a) SEM image and b) TEM image of UiO-66. c) FTIR spectra of BTA, UiO-66, and BU. d) SEM image and e) TEM image of BU. f) N_2 adsorption/desorption isotherms of UiO-66 and BU. g) XRD patterns, h) high-resolution XPS spectra (Zr 3d), i) high-resolution XPS spectra (Si 2p), and j) TGA curves of UiO-66 and BU.

structures with a size of 20–50 nm, consistent with those in the literature.^[34] FTIR spectra show BTA and BU have bands at 980 and 1096 cm^{-1} attributed to the stretching vibration of two adjacent C–N bonds on the benzene ring of BTA (Figure 3c).^[35] The band at 1100 cm^{-1} is attributed to the stretching vibration of Si–OH.^[36] The facts indicate that BTA was successfully loaded and KH570 was grafted.

The nitrogen sorption experiments show each sample had standard type-I form isothermal adsorption curves (based on IUPAC definition) with a very significant uptake at relatively low pressure, suggesting that each sample had a microporous window (Figure 3f).^[35,37] The Brunauer–Emmett–Teller (BET) surface area of UiO-66 was estimated to be 593.9 $\text{m}^2 \text{g}^{-1}$, and the pore volume was 1.191 $\text{cm}^3 \text{g}^{-1}$. It was significantly higher than that of BU (BET surface area = 364.4 $\text{m}^2 \text{g}^{-1}$ and pore volume = 0.197 $\text{cm}^3 \text{g}^{-1}$) because of the loading of BTA, which blocks the pores and reduces the surface area.^[35] XRD pattern of BU was similar to that of UiO-66 (Figure 3g) with the same crystallinity and characteristic peaks at $2\theta = 7.34^\circ$ and 8.51° , indicating that

the morphology and surface crystallinity of the MOF slightly changed after loading with BTA and grafting with KH570.^[34] Furthermore, the Zr 3d high-resolution spectrum (Figure 3h) exhibited two peaks at 180 and 182.1 eV attributed to Zr 3d_{5/2} and Zr 3d_{3/2}, respectively, indicating the successful preparation of MOF. The increase in intensity of Si 2p also indicates the grafting of KH570 (Figure 3i). The significant weight loss of BU from 190 to 285 °C corresponded to the weight loss of the loaded BTA.^[30] BU had a decreased residual weight of 44.1 wt%, much less than that of UiO-66 nanospheres (52.2 wt%) further indicating the loading of BTA (Figure 3j).

MUX coatings were fabricated by incorporating PMVS20 with BU. As shown in Figure 4a, the tensile tests show that the strain corresponding to the ultimate stress of the coating was enhanced by the addition of BU, resulting from the crosslinking between MOF and polymeric networks. Excessive BU would increase the crosslink density and hence decrease the deformability of the coating. As BU content increased, the adhesion strength significantly increased (Figure 4b). This is because siloxane covalent

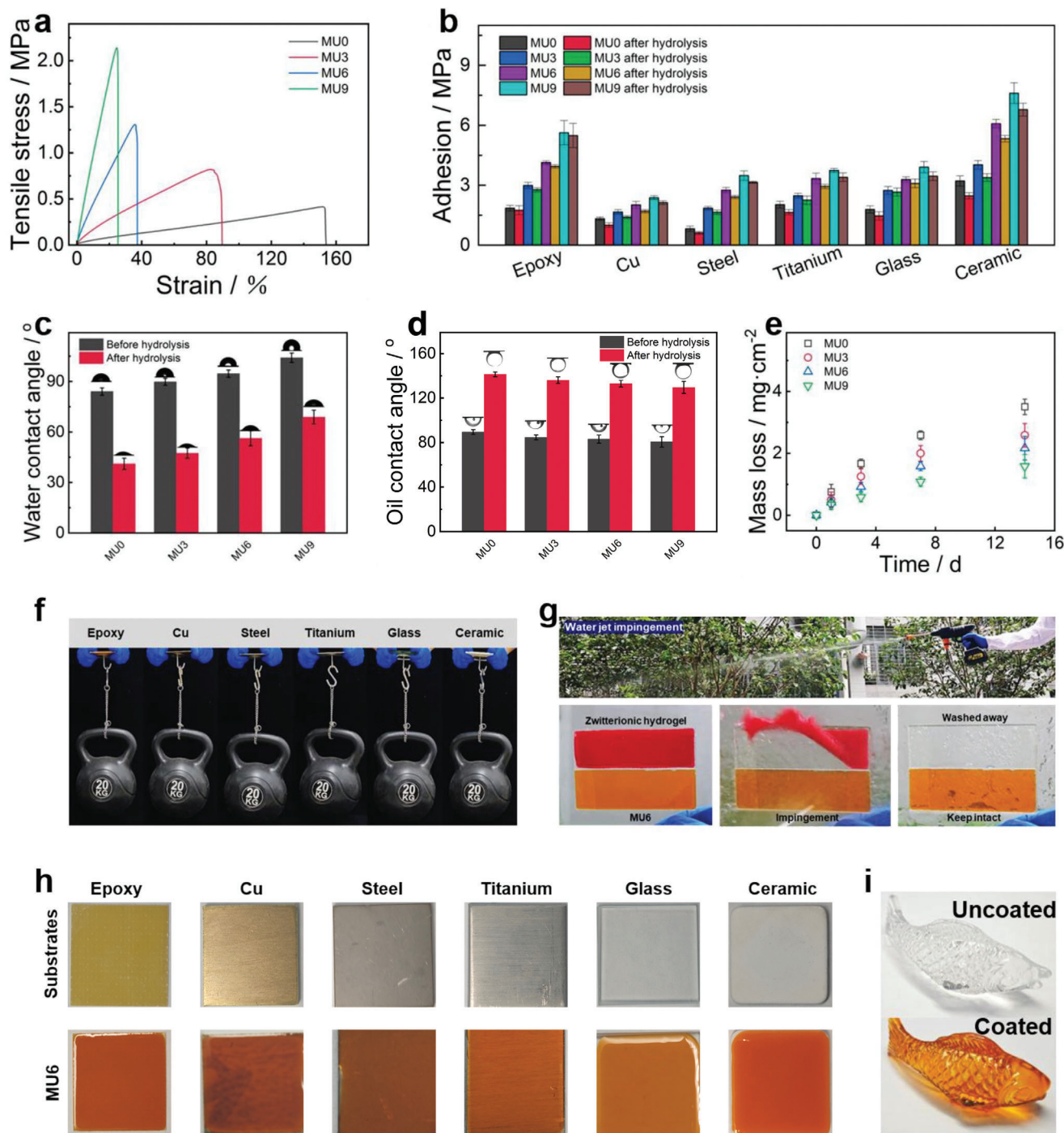


Figure 4. Characterizations of MUx. a) Tensile curves of MUx. b) The adhesion strength of MUx before and after hydrolysis. c) Water contact angles and d) oil contact angles of MUx before and after hydrolysis. e) Variation in the mass loss of MUx. f) MU6 bonded to substrates with a 20kg kettlebell load. g) Water jet impingement test of MU6 (water pressure ≈ 2 MPa). h) MU6 coated on various substrates. i) MU6-coated glass fish.

bonds between the coating and substrate increased.^[18] MU9 possessed the highest adhesion strength to epoxy (5.63 ± 0.06 MPa), copper (Cu) (2.38 ± 0.10 MPa), stainless steel (3.48 ± 0.24 MPa), titanium (3.75 ± 0.10 MPa), glass (3.91 ± 0.26 MPa), and ceramic (7.61 ± 0.52 MPa) before hydrolysis. However, the adhesion strength of each coating slightly decreased after hydro-

lysis. This is understandable because the hydrolysis occurred only on the top layer of the coating while the bottom layer remained unhydrolyzed. The adhesion strength of each BU-doped coating was higher than 1.5 MPa. Accordingly, these coatings have higher adhesion than the conventional FRC and SPC coatings, so they can be used in marine environments. The water contact angle

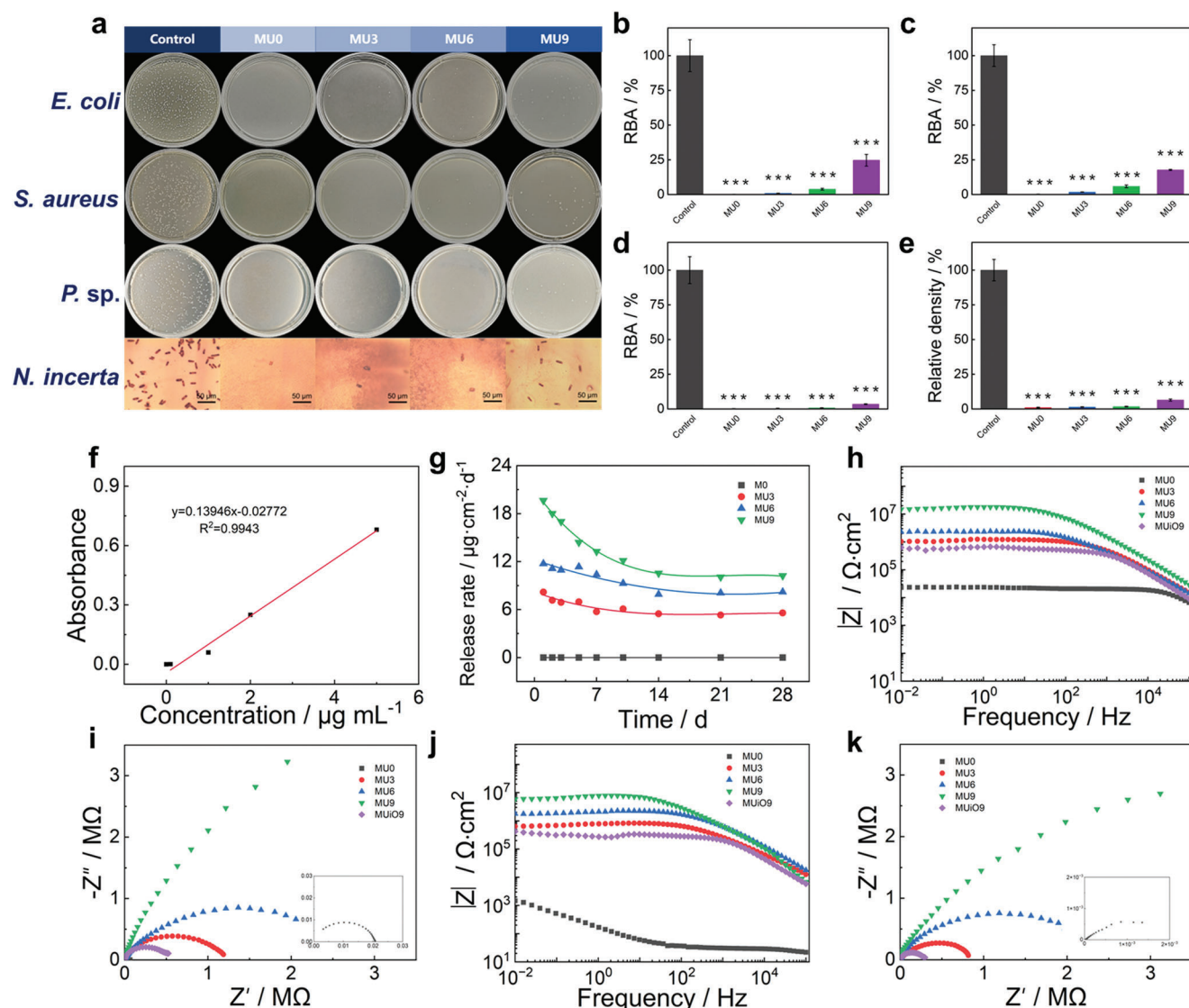


Figure 5. Anti-biofouling and anti-corrosion properties of MUX. a) Photographs of agar plates of bacterial colonies *Escherichia coli* (*E. coli*), *Staphylococcus aureus* (*S. aureus*), and *Pseudomonas* sp. (*P. sp.*) and microscopy images of MUX coatings after incubation with diatom (*N. incerta*). b–d) Relative bacterial adsorption (RBA) of *E. coli*, *S. aureus*, and *P. sp.*, respectively. e) Relative density of *N. incerta*. $n = 3$, biologically independent samples. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. f) The absorbance-concentration standard curve of BTA. g) Time dependence of the release rate of BTA from MUX in ASW at 25 °C. h) Evolution of Bode plots and i) Nyquist plots of MUX and MUIO9 for 4 days. j) Evolution of Bode plots and k) Nyquist plots of MUX and MUIO9 in 7 days.

slightly increased whereas the oil contact angle slightly decreased with BU content after hydrolysis (Figure 4c,d). This is attributed to the increase in hydrophobicity of BU.

Figure 4e shows the time dependence of the mass loss of MUX coatings in ASW. Generally, each film started to lose weight after immersing in ASW due to the hydrolysis of zwitterion precursors, the release of BTA, and the hydrolytic degradation of the polyester segments. Compared to non-degradable coatings such as FRC and SPC coatings, MUX coating can continually renew its surface and effectively inhibit the adhesion of inorganic substances and dead bacteria in static conditions. The mass loss decreased with BU content because the content of zwitterion or polyester segments decreased. As a result, the crosslinking den-

sity increased and the hydrolytic degradation rate decreased. As BU content increased, the crosslink density increased and the zwitterions content decreased, leading to a reduction in the water absorption rate (Figure S4, Supporting Information).

MU6 with proper mechanical properties and hydrolytic degradation rate was chosen throughout the test. As shown in Figure 4f, MU6-bonded substrates could lift a 20 kg kettlebell. The water impact resistance of the coating was evaluated (Figure 4g and Movie S1, Supporting Information). MU6 and zwitterionic hydrogel coating without self-generating ability were subjected to a water jet with a pressure of ≈ 2 MPa. The latter was readily washed away by the water flow, whereas MU6 stayed in place upon the water flow. As shown in Figure 4h,i, the precursor

of MU6 coating can be widely applied to various substrates without any surface treatment. Namely, MU6 exhibited robust and consistent adhesion strength, rendering it suitable for utilization on an array of substrates.

2.3. Anti-Biofouling and Anti-Corrosion Performances

Bacteria play an essential role in the formation of biofilms in marine biofouling. Gram-positive bacteria *Staphylococcus aureus* (*S. aureus*), gram-negative bacteria *Escherichia coli* (*E. coli*), and marine bacteria *Pseudomonas* sp. (*P. sp.*) were used as a model for antibacterial assays because of their widespread presence in marine environments. The antibacterial capability was directly visualized by typical optical photographs of agar plates dispersed with bacterial suspension after incubation with MUx coatings (Figure 5a). Figure 5b–d show the relative bacterial adsorption (RBA) of the bacteria. A large number of bacterial colonies were observed on the plates of the control group. In contrast, there were almost no bacterial colonies on the MU0 plate, indicating its ability to inhibit all the bacteria ($p < 0.001$ for all bacteria) due to the broad-spectrum antimicrobial activity of the zwitterions. Note that the number of bacterial colonies slightly increased with BU content when it was less than 9%, but some bacterial colonies were observed on the MU9 coating surface in that the increase in BU content decreased the concentration of the zwitterions. Anyhow, BU content should be limited to $< 9\%$.

Diatom (*N. incerta*) was used to further examine the anti-biofouling diversity of MUx coatings. After 24 h of immersion in the diatom suspension under daylight/dark illumination cycles, a large number of diatoms were observed in the control since the diatoms like hydrophobic surface (Figure 5a).^[18] The diatom attachment inhibition was tested by the amounts of settled diatoms on MUx coatings. As illustrated in Figure 5e, as BU content increased from 0 wt% to 9 wt%, the relative density of diatoms on the coating surface increased from $\approx 0.95\%$ to $\approx 6.38\%$ ($p < 0.001$). A similar phenomenon was also observed in the antibacterial assays. The facts indicate the zwitterions in MUx coating can effectively inhibit biofouling.

On the other hand, the controlled release of corrosion inhibitors would improve the anti-corrosion ability and prolong the anti-corrosion duration of the coating. Figure 5g shows the time dependence of BTA release rate from MUx coatings in ASW at 25 °C relative to the standard curve of BTA (Figure 5f and S5, Supporting Information). For each BU-doped coating, the release rate was relatively high at the initial stage but then gradually decreased and leveled off in 14 days. The stable BTA release resulted from the continuous degradation and hydrolysis of the coating. The constant release rates were ≈ 5 , 8, and $11 \mu\text{g cm}^{-2} \text{d}^{-1}$ for MU3, MU6, and MU9, respectively. The release rate increases with BU content. Namely, the degradation of polyester segments and hydrolysis of zwitterion precursors enable the prolonged release of BTA, which can be controlled by regulating BU content.

The anti-corrosion performance of MUx coating was evaluated by electrochemical measurements after immersion in ASW for 4 days and 7 days. To clarify the impact of MOF, a coating doped with UiO-66 but without BTA (MUiO9) was also prepared. In terms of the Bode diagram (Figure 5h), MUiO9 coating exhibited a markedly higher low-frequency impedance modulus ($|Z|_{0.01}$

Hz) of $6.3 \times 10^5 \Omega \text{ cm}^2$ than MU0 coating ($2.3 \times 10^4 \Omega \text{ cm}^2$), indicating that UiO-66 can effectively inhibit corrosion. This is because the porous nanostructure of UiO-66 has effective corrosion resistance.^[29,38,39] The BU-doped coatings had characteristic low-frequency impedance moduli ($|Z|_{0.01} \text{ Hz}$) of 1.0×10^6 , 2.3×10^6 , and $14.8 \times 10^6 \Omega \text{ cm}^2$, much higher than those of MU0 and MUiO9 coatings, indicating that the incorporation of BTA further enhanced the corrosion resistance.^[40,41] A similar trend can be observed in the Nyquist plots, where the capacitance semicircle diameter of each BU-doped coating was larger than those of MU0 and MUiO9. Moreover, the diameter increased with BU content (Figure 5i). The synergistic effect of UiO-66 and the constant release of BTA due to the continuous degradation and hydrolysis of the coating efficiently impeded the corrosion process.^[26,38] This is further corroborated by the long-term tests (Figure 5j,k). The phase angle diagram of BU-doped coating exhibited a single peak, that is, it had a single time constant, indicating that the corrosive medium was impeded from reaching the metal surface owing to the notable barrier properties conferred by the coating (Figure S6a,b, Supporting Information).

3. Conclusion

In summary, we present a self-regenerating zwitterionic hydrogel coating with BTA-loaded MOF for integrated anti-biofouling and anti-corrosion. The silanol condensation between PMVSx and BU enables the coating to have good mechanical properties and to build a robust adhesion to substrates. The coating hydrolyzes and degrades in water to create a self-renewing zwitterionic dynamic surface, effectively resisting the adhesion of bacteria (*Escherichia coli*, *Staphylococcus aureus*, and *Pseudomonas* sp.) and diatom (*Navicula incerta*). In addition, MOF (UiO-66) is disintegrated in alkaline and saline environments, enabling the continuous release of BTA and hence significantly improving the anti-corrosion performance. This study provides a strategy to develop hydrogel coating with good mechanical properties as well as high anti-biofouling and anti-corrosion performances.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

anti-biofouling, anti-corrosion, hydrogel coating, metal-organic framework, zwitterions

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- [1] P.-Y. Qian, A. Cheng, R. Wang, *Nat. Rev. Microbiol.* **2022**, 20, 671.
- [2] H. Qiu, K. Feng, A. Gapeeva, K. Meurisch, S. Kaps, X. Li, L. Yu, Y. Kumar, R. Adelung, M. Baum, Y. K. Mishra, R. Adelung, M. Baum, *Prog. Polym. Sci.* **2022**, 127, 101516.
- [3] H. Jin, L. Tian, W. Bing, J. Zhao, L. Ren, *Prog. Mater. Sci.* **2022**, 124, 100889.
- [4] X. Zang, Y. Ni, Q. Wang, Y. Cheng, J. Huang, *Mater. Today* **2024**, 75, 210.
- [5] J. Yang, B. Xue, Y. Zhou, M. Qin, W. Wang, Y. Cao, *Adv. Mater. Technol.* **2021**, 6, 2000911.
- [6] D. C. Leslie, A. Waterhouse, J. B. Berthet, T. M. Valentin, A. L. Watters, A. Jain, P. Kim, B. D. Hatton, A. Nedder, K. Donovan, E. H. Super, C. Howell, C. P. Johnson, T. L. Vu, D. E. Bolgen, S. Rifai, A. R. Hansen, M. Aizenberg, M. Super, J. Aizenberg, D. E. Ingber, *Nat. Biotechnol.* **2014**, 32, 1134.
- [7] I. Amara, W. Miled, R. Ben Slama, N. Ladhari, *Environ. Toxicol. Pharmacol.* **2018**, 57, 115.
- [8] W. J. Yang, K. G. Neoh, E. T. Kang, S. L. M. Teo, D. Rittschof, *Prog. Polym. Sci.* **2014**, 39, 1017.
- [9] P. Hu, Q. Xie, C. Ma, G. Zhang, *Langmuir* **2020**, 36, 2170.
- [10] M. Liu, S. Li, H. Wang, R. Jiang, X. Zhou, *Polym. Chem.* **2021**, 12, 3702.
- [11] F. Koschitzki, O. Özcan, R. Wanka, M. Krisam, A. Rosenhahn, *Adv. Mater. Interfaces* **2024**, 11, 2400370.
- [12] X. Yao, J. Liu, C. Yang, X. Yang, J. Wei, Y. Xia, X. Gong, Z. Suo, *Adv. Mater.* **2019**, 31, 1903062.
- [13] Q. Li, C. Wen, J. Yang, X. Zhou, Y. Zhu, J. Zheng, G. Cheng, J. Bai, T. Xu, J. Ji, S. Jiang, L. Zhang, P. Zhang, *Chem. Rev.* **2022**, 122, 17073.
- [14] J. Liu, S. Qu, Z. Suo, W. Yang, *Natl. Sci. Rev.* **2021**, 8, nwaa254.
- [15] H. Yuk, T. Zhang, S. Lin, G. A. Parada, X. Zhao, *Nat. Mater.* **2016**, 15, 190.
- [16] Y. Yu, H. Yuk, G. A. Parada, Y. Wu, X. Liu, C. S. Nabzdyk, K. Youcef-Toumi, J. Zang, X. Zhao, *Adv. Mater.* **2019**, 31, 1807101.
- [17] Z. Yang, Y. He, S. Liao, Y. Ma, X. Tao, Y. Wang, *Sci. Adv.* **2021**, 7, eabf9117.
- [18] Y. Zhang, Q. Xie, J. Pan, C. Ma, G. Zhang, *Chem. Eng. J.* **2024**, 485, 149959.
- [19] G. Dai, Q. Xie, X. Ai, C. Ma, G. Zhang, *ACS Appl. Mater. Interfaces* **2019**, 11, 41750.
- [20] J. Pan, L. Mei, H. Zhou, C. Zhang, Q. Xie, C. Ma, *Prog. Org. Coatings* **2022**, 163, 106674.
- [21] Z. Yu, X. Li, X. Li, B. Zheng, D. Li, D. Xu, *Adv. Funct. Mater.* **2023**, 33, 2305995.
- [22] Q. Xie, J. Pan, C. Ma, G. Zhang, *Soft Matter* **2019**, 15, 1087.
- [23] J. Pan, Q. Xie, H. Chiang, Q. Peng, P. Y. Qian, C. Ma, G. Zhang, *ACS Sustainable Chem. Eng.* **2020**, 8, 1671.
- [24] G. Cai, P. Yan, L. Zhang, H. C. Zhou, H. L. Jiang, *Chem. Rev.* **2021**, 121, 12278.
- [25] H. Zheng, Y. Zhang, L. Liu, W. Wan, P. Guo, A. M. Nyström, X. Zou, *J. Am. Chem. Soc.* **2016**, 138, 962.
- [26] W. Cai, J. Wang, C. Chu, W. Chen, C. Wu, G. Liu, *Adv. Sci.* **2019**, 6, 1801526.
- [27] M. Ramezanzadeh, B. Ramezanzadeh, G. Bahlakeh, A. Tati, M. Mahdavian, *Chem. Eng. J.* **2021**, 408, 127361.
- [28] V. Kavimani, S. P. K., R. Rajesh, D. Rammasamy, N. Babu, T. Yang, B. Prabakaran, S. Jothi, *Appl. Surf. Sci.* **2017**, 424, 63.
- [29] Y. Zhang, J. Wang, S. Zhao, M. Serdechnova, C. Blawert, H. Wang, M. L. Zheludkevich, F. Chen, *ACS Appl. Mater. Interfaces* **2021**, 13, 51685.
- [30] J. Zhou, H. Liu, Y. Sun, K. Chen, *Adv. Mater. Interfaces* **2022**, 9, 2102001.
- [31] L. Mei, X. Ai, C. Ma, G. Zhang, *J. Mater. Chem. B* **2020**, 8, 5434.
- [32] D. Sun, P. R. Adiyala, S. J. Yim, D. P. Kim, *Angew. Chemie. Int. Ed.* **2019**, 58, 7405.
- [33] D. Bůžek, S. Adamec, K. Lang, J. Demel, *Inorg. Chem. Front.* **2021**, 8, 720.
- [34] J. Du, L. Chen, X. Zeng, S. Yu, W. Zhou, L. Tan, L. Dong, C. Zhou, J. Cheng, *ACS Appl. Mater. Interfaces* **2020**, 12, 28576.
- [35] A. Helal, M. Nasiruzzaman Shaikh, M. Abdul Aziz, *J. Photochem. Photobiol. A Chem.* **2020**, 389, 112238.
- [36] Y. Kuai, F. Wang, J. Yang, Z. Xu, H. Li, X. Xu, Y. Nuli, J. Wang, *ACS Appl. Mater. Interfaces* **2021**, 13, 26971.
- [37] Y. Zhao, F. Jiang, Y. Q. Chen, J. M. Hu, *Corros. Sci.* **2020**, 168, 108563.
- [38] X. Zhang, Z. Chen, X. Liu, S. L. Hanna, X. Wang, R. Taheri-Ledari, A. Maleki, P. Li, O. K. Farha, *Chem. Soc. Rev.* **2020**, 49, 7406.
- [39] S. Qiu, T. Y. Sun, Y. Su, L. F. Huang, H. Zhao, Q. Xue, L. Wang, *Chem. Eng. J.* **2021**, 426, 131879.
- [40] I. I. Udoh, O. O. Ekerenam, E. F. Daniel, A. I. Ikeuba, D. I. Njoku, S. K. Kolawole, I. I. N. Etim, W. Emori, C. N. Njoku, I. P. Etim, P. C. Uzoma, *Adv. Colloid Interface Sci.* **2024**, 330, 103209.
- [41] X. Li, X. Zhu, A. Feng, M. An, P. Liu, Y. Zu, *J. Mater. Res. Technol.* **2024**, 29, 5667.